

KAIMANA UTAROM, Indonesia

WMO: 977600

Lat: 3.6439S

Lon: 133.6969E

Elev: 3

StdP: 101.29

Time Zone: 9.00 (E09)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.8	23.2	20.6	15.2	25.2	21.1	15.8	25.3	8.3	26.2	7.3	26.9	0.8	90	0.337

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	6.4	33.0	26.9	32.4	26.7	31.9	26.6	27.5	31.6	27.2	31.2	27.0	30.9	3.1	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.3	21.7	29.8	26.1	21.5	29.6	25.8	21.1	29.3	87.4	31.9	86.1	31.6	85.0	31.1	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.7	5.7	5.1	DB	21.3	35.6	2.8	2.0	19.3	37.1	17.6	38.2	16.1	39.3	14.1	40.8	
			WB	20.2	28.9	2.5	0.9	18.4	29.5	16.9	30.1	15.5	30.6	13.7	31.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	28.3	28.2	28.2	27.9	27.6	26.9	26.2	26.1	26.5	27.4	27.9	28.4
	DBStd	1.21	1.09	1.11	1.04	0.80	0.79	0.89	0.78	0.82	0.81	0.71	0.83	0.99
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6373	568	510	564	537	546	506	503	498	496	539	537	570
	CDD18.3	3332	309	277	306	287	287	256	245	239	246	281	287	312
	CDH23.3	33829	3456	3056	3325	3067	3058	2422	2011	1835	2156	2881	3097	3467
CDH26.7	9873	1252	1103	1161	1021	932	486	224	176	363	809	1071	1274	
Wind	WSAvg	2.0	2.3	2.2	2.1	1.5	1.4	2.0	2.6	2.9	2.1	1.6	1.3	1.9
Precipitation	PrecAvg	2399	226	209	245	264	288	187	141	165	116	143	231	194
	PrecMax	3207	410	328	563	493	586	581	476	322	372	361	336	280
	PrecMin	1472	77	104	100	106	69	40	20	13	6	25	76	95
	PrecStd	554	93	76	115	108	119	134	123	88	97	102	76	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.5	33.5	33.4	32.6	31.8	31.0	29.6	29.4	30.9	31.8	33.0	33.5
		MCWB	27.2	27.1	27.1	26.9	26.8	26.8	25.7	25.6	25.9	26.2	26.8	27.2
	2%	DB	32.6	32.6	32.5	31.9	31.2	30.2	29.0	28.7	30.1	31.1	32.0	32.6
		MCWB	26.7	26.7	26.9	26.7	26.6	26.3	25.5	25.2	25.8	26.1	26.6	26.9
	5%	DB	31.8	31.8	31.7	31.2	30.7	29.7	28.5	28.2	29.2	30.5	31.4	32.0
		MCWB	26.5	26.5	26.6	26.5	26.5	26.0	25.2	25.0	25.6	26.0	26.4	26.7
	10%	DB	31.1	31.1	31.0	30.7	30.2	29.1	27.9	27.7	28.6	30.0	30.8	31.3
		MCWB	26.3	26.1	26.2	26.3	26.3	25.8	24.9	24.8	25.3	25.9	26.2	26.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.7	27.6	27.7	27.6	27.4	27.2	26.2	26.1	26.6	27.0	27.6	27.7
		MCDB	32.3	32.4	31.9	31.3	30.8	29.9	28.5	28.6	29.5	30.6	32.3	32.0
	2%	WB	27.2	27.1	27.2	27.2	27.1	26.6	25.7	25.6	26.1	26.6	27.1	27.2
		MCDB	31.4	31.7	31.3	30.9	30.3	29.5	28.2	28.0	29.1	30.0	31.0	31.4
	5%	WB	26.8	26.7	26.8	26.9	26.7	26.2	25.5	25.2	25.7	26.3	26.7	27.0
		MCDB	30.7	30.9	30.8	30.4	29.8	29.0	27.8	27.5	28.5	29.6	30.3	31.1
	10%	WB	26.5	26.4	26.5	26.6	26.5	25.9	25.2	25.0	25.4	26.1	26.5	26.7
		MCDB	30.3	30.3	30.2	29.9	29.5	28.6	27.4	27.3	28.0	29.2	29.9	30.5
Mean Daily Temperature Range	5% DB	MDBR	6.2	6.5	6.3	6.1	5.4	4.1	3.5	3.3	4.1	5.4	6.4	6.4
		MCDBR	7.0	7.1	7.1	6.7	6.1	4.9	4.1	3.9	5.3	6.4	7.1	7.1
		MCWBR	2.9	2.8	2.8	2.8	2.6	2.3	2.0	2.0	2.3	2.7	2.8	2.8
	5% WB	MCDBR	6.5	6.9	6.6	6.2	5.6	4.6	3.9	3.7	4.9	5.8	6.6	6.6
		MCWBR	2.9	2.9	2.8	2.7	2.5	2.3	2.1	2.0	2.3	2.6	2.8	2.8
Clear-Sky Solar Irradiance	taub		0.400	0.399	0.396	0.397	0.409	0.419	0.419	0.426	0.434	0.457	0.440	0.414
	taud		2.533	2.528	2.538	2.543	2.510	2.499	2.495	2.463	2.438	2.343	2.401	2.484
	Ebn at Noon		935	935	927	902	866	843	849	864	879	872	890	918
	Edh at Noon		110	111	109	105	104	103	104	112	118	132	124	114
All-Sky Solar Radiation	RadAvg		5.11	5.24	5.15	5.03	4.83	4.33	4.17	4.43	4.85	5.29	5.30	5.07
	RadStd		0.27	0.24	0.18	0.23	0.24	0.28	0.25	0.25	0.28	0.28	0.22	0.24

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page